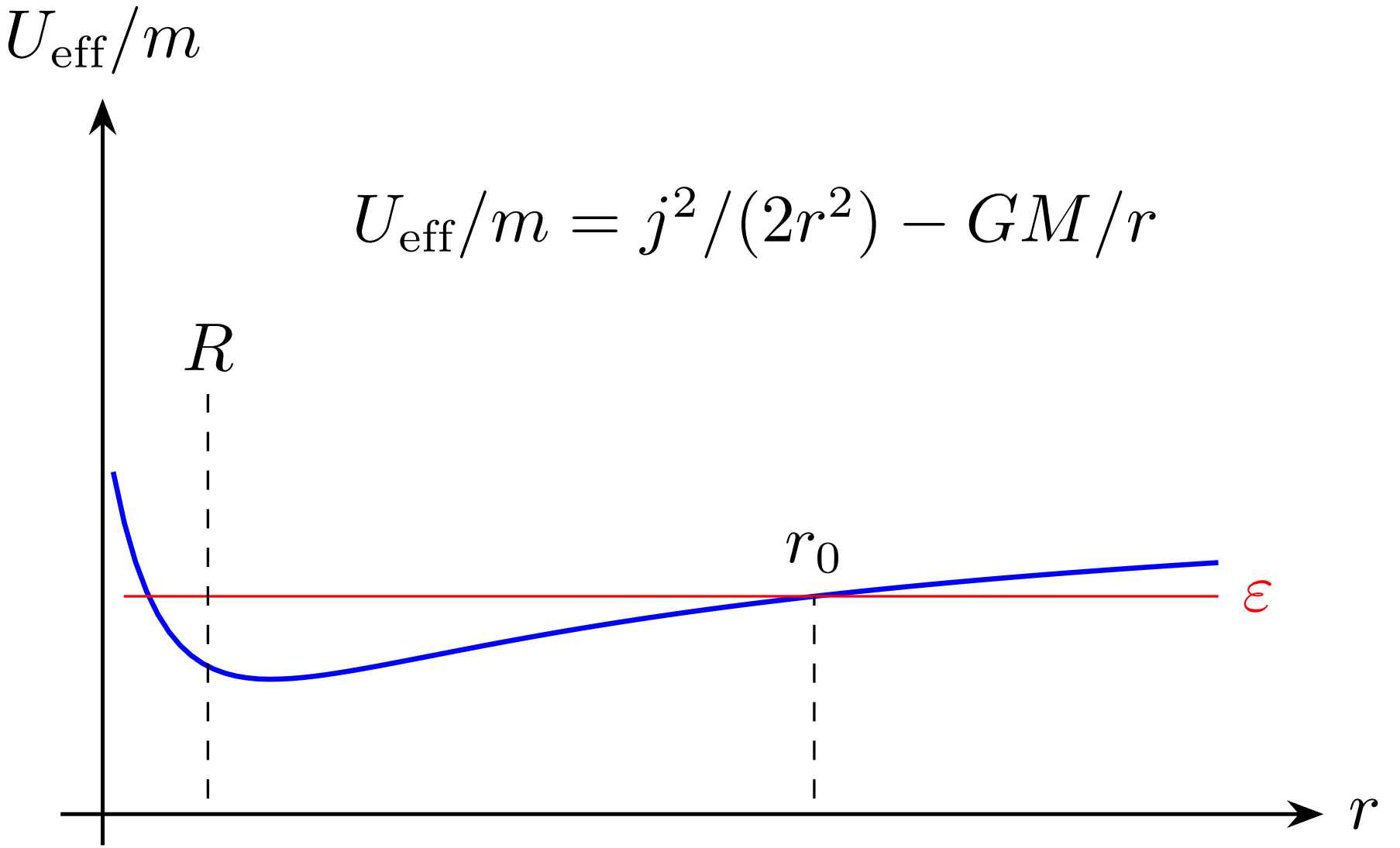
U_{eff}/m
$U_{\text{eff}}/m = j^2/(2r^2) - GM/r$
R
r_0
$\mathcal{E}$
r